Shortfin squid (*Illex illecebrosus*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** High
- **Vulnerability:** Low

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1	3	
Complexity in Reproductive Strategy	1.52	2.7	
Dispersal of Early Life Stages	1.32	3	
Egg and Larval Survival and Settlement Requirements	2.88	2.3	
Historical Variability in Temperature	2.04	2.6	
Juvenile Habitat Requirements	1.04	3	
Other Stressors	1.44	2.4	
Population Growth Rate	1.92	2.1	
Prey Specificity	1.56	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.96	2.6	
Spawning Cycle	2.08	2	
Stock Size/Status	2.24	1.6	
Total Sensitivity | Certainty	**Low (1) | 0.89**		

Exposure

Exposure Factors	Mean	SD	Tally
Sea Surface Temperature	3.14	1.09	
Sea Surface Salinity	3.12	0.86	
Surface pH	3.85	0.24	
Mixed layer depth (delta rho = 0.03)	1.05	0.24	
Small zooplankton integrated biomass	1.37	0.40	
Medium zooplankton integrated biomass	1.58	0.48	
Large zooplankton integrated biomass	1.17	0.24	
Total Exposure	**High (3.13)**	**0.48**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (3.13)**	**0.48**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Shortfin squid. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Low	None
Exposure	High	High	None
Vulnerability	Low	Low	None
Directional Effect	Positive	Neutral	Decrease

Stocks

Shortfin squid is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

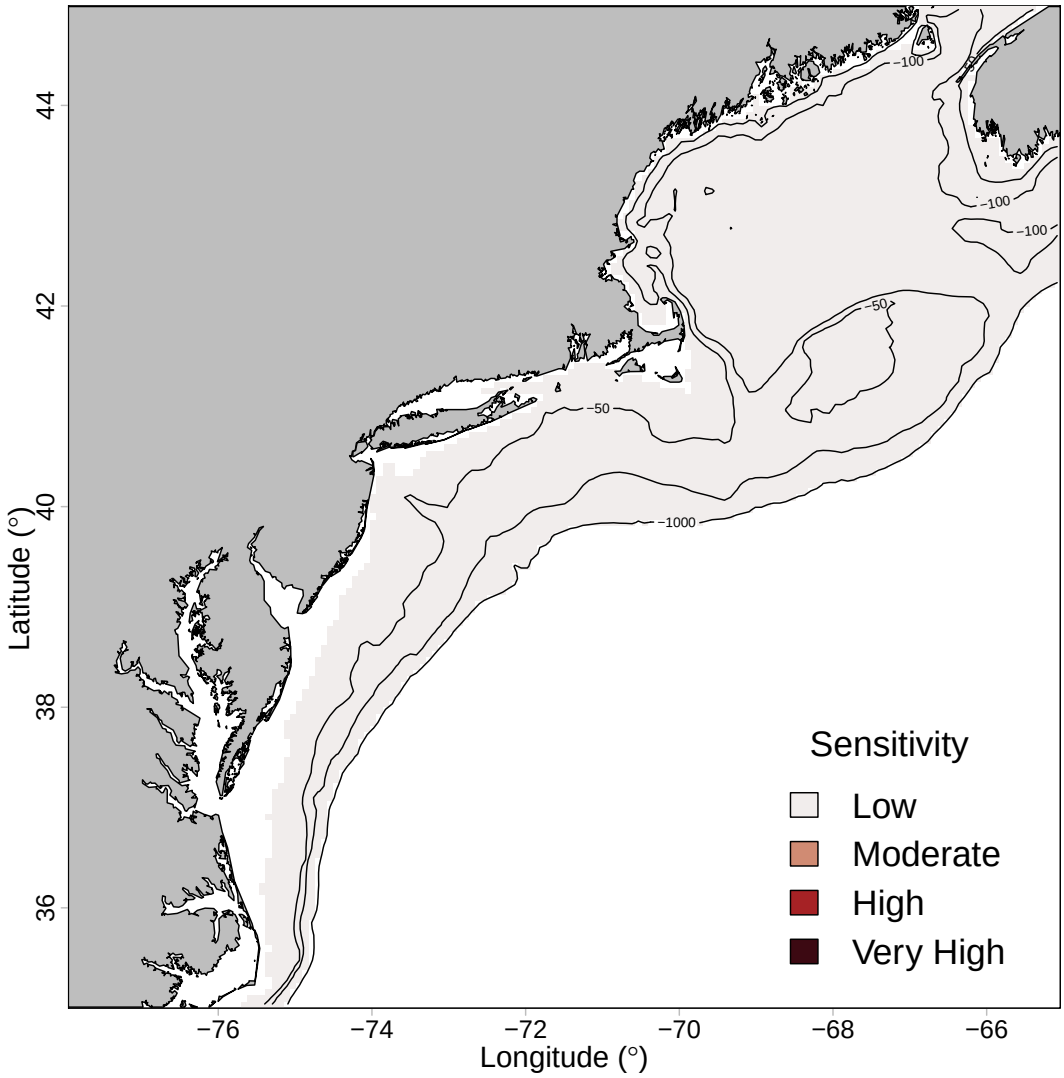


Figure 2: Shortfin squid Sensitivity on the Northeast Shelf

Short-finned squid received a **Low** sensitivity score, with a certainty of 89%. Only one attribute – Egg and Larval Survival and Settlement Requirements (2.9) – received a score greater than 2.5 (Figure 1). Short-finned squid rely on the Gulf Stream for transport between southern spawning habitats and northern feeding grounds. Therefore, recruitment has been linked to climate oscillations such as the North Atlantic Oscillation that drive shifts in the Gulf Stream (Dawe, Colbourne, and Drinkwater 2000) (O'Dor and Dawe 2013) (Hendrickson and Holmes 2004).

In the previous Climate Vulnerability Assessment for the Northeast Shelf, Short-finned squid also received a Low sensitivity score. In that analysis, no attributes had a score greater than 2.5 (Hare et al. 2016).

Exposure:

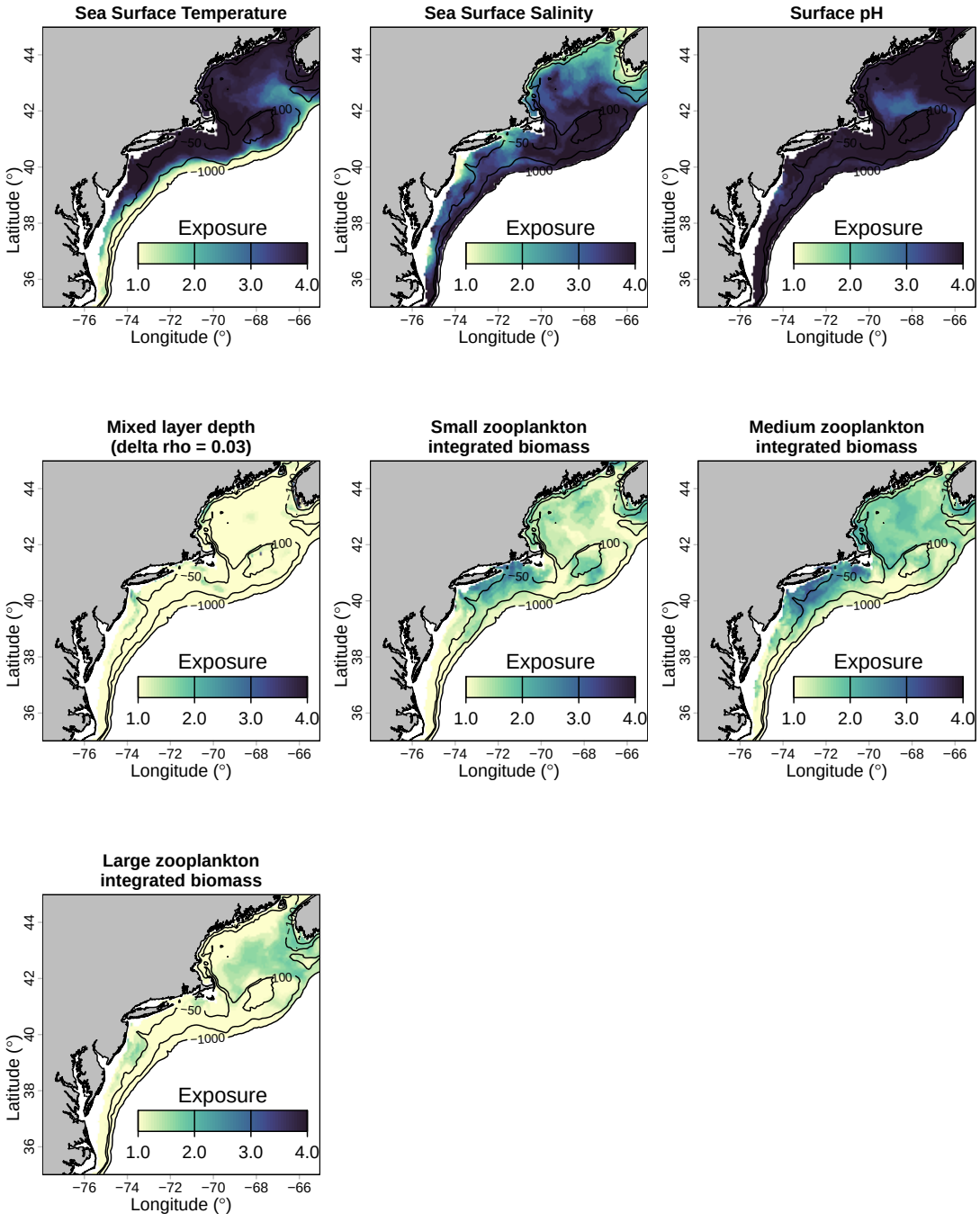


Figure 3: Maps of each Exposure Factor for Shortfin squid. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure for shortfin squid was **High** (3.13 +/- 0.48) (Figure 1). Exposure was consistently High throughout the Northeast Shelf, with Very High scores off the coast of Cape Cod and over Georges

Bank (Figure 4). These patterns were driven by high sea surface temperature, salinity, and pH scores (Figure 3).

Shortfin squid received the same exposure score in the previous Assessment (Hare et al. 2016).

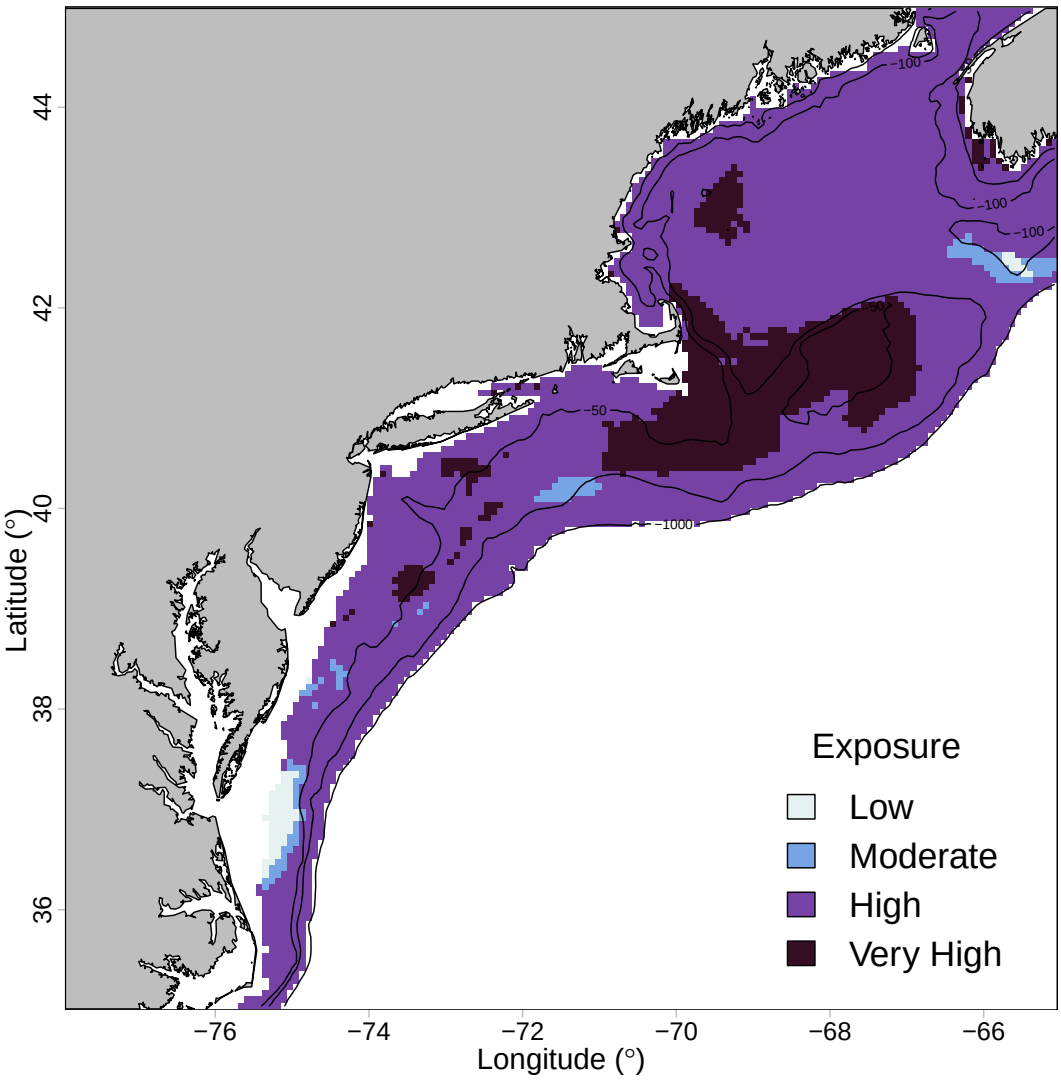


Figure 4: Total Exposure on the Northeast Shelf for Shortfin squid. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

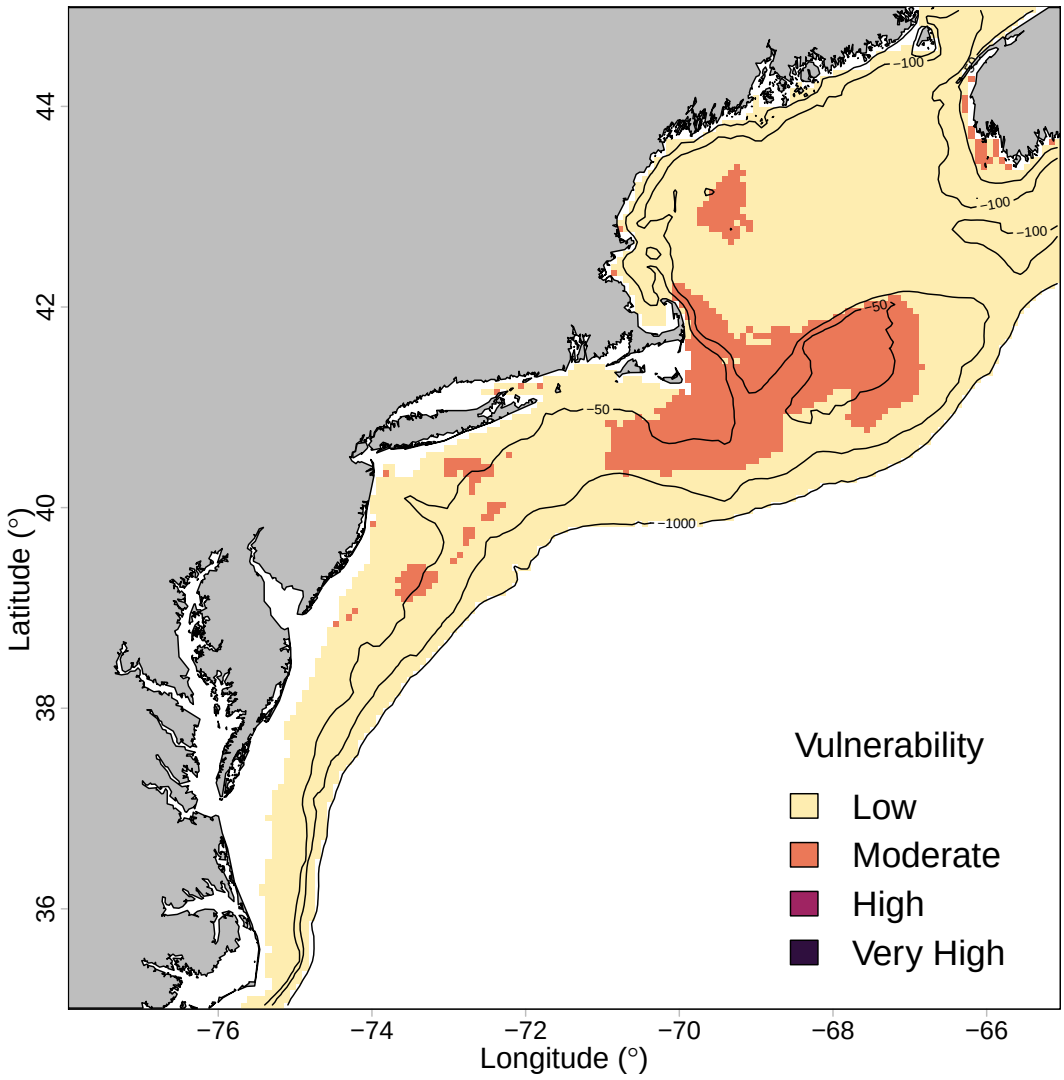


Figure 5: Vulnerability on the Northeast Shelf for Shortfin squid. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean overall Vulnerability rank for shortfin squid was **Low** (3.13 +/- 0.48) (Figure 1). Vulnerability was Moderate south of Cape Cod and on Georges Bank, but Low throughout the rest of the Northeast Shelf (Figure 5).

Shortfin squid also received a Low Vulnerability score in the previous assessment (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

85% of data quality scores were greater than or equal to 2, indicating strong data quality. Only Stock Size/Status was considered a data gap (mean attribute data quality score < 2) (Figure 1).

Model Confidence Score:

The shortfin squid ensemble model received a model confidence score of 2 (+/- 0.4), indicating moderate confidence in the model. The predicted distributions aligned with expert expectations. The model also captured the recent increase in shortfin presence in the Gulf of Maine. The ensemble model AUC score was 0.83, indicating good model performance.

Additional Metrics

Distribution Shifts:

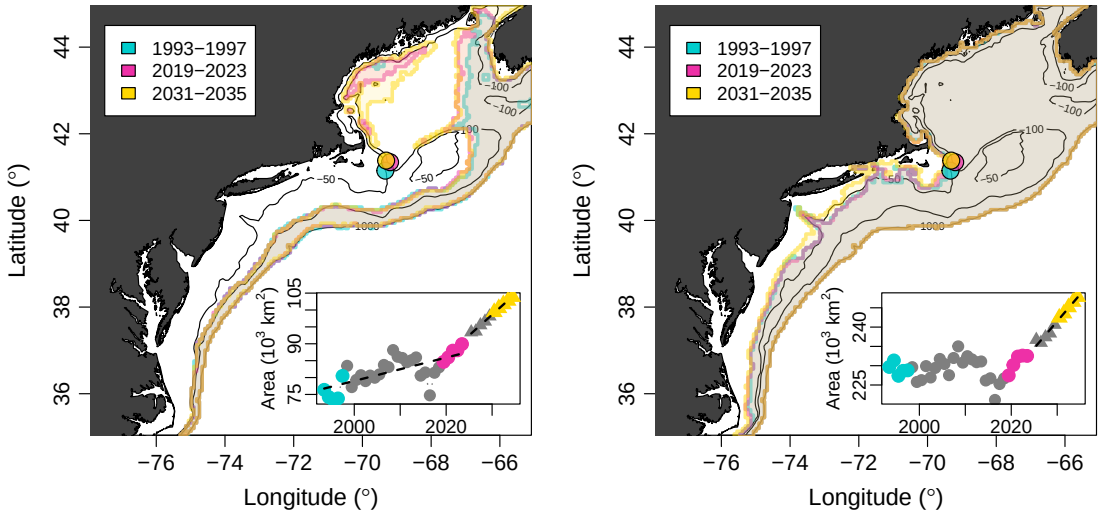


Figure 6: Distribution shifts on the Northeast Shelf for Shortfin squid. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Core habitat for shortfin squid significantly increased between 1993 and 2023. Shortfin squid core habitat expanded into the Gulf of Maine during this time (Figure 6). The weighted centroid shifted 25 km northeast.

Under near future climate conditions, these trends are predicted to continue. Both shortfin squid core habitat and their range are predicted to significantly increase. Core habitat use in the coastal Gulf of Maine will increase, and their overall range is expected to shift towards the coast (Figure 6). As a result, the weighted centroid is predicted to shift 10 km west-northwest.

In the previous assessment, shortfin squid received a Very High Distributional Vulnerability rank due to their high mobility, large migrations, and broad larval dispersal (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	76,082	229,374	7.05	9.06
2019-2023	24.6 km NE	87,505 (+11,423)	231,193 (+1,819)	6.92 (-0.12)	8.98 (-0.08)
2031-2035	10.4 km WNW	101,683 (+14,177)	245,237 (+14,043)	6.98 (+0.06)	9.06 (+0.08)

Directional Effect:

Experts predicted that climate change will have a **neutral** effect on short-finned squid. Certainty was 96%. Recruitment is highly dependent on transport mechanisms such as the Gulf Stream and catch per unit effort is correlated with cooler bottom temperatures (Salois et al. 2023) (Dawe, Colbourne, and Drinkwater 2000) (O'Dor and Dawe 2013) (Hendrickson and Holmes 2004). However, short-finned squid are able to adapt to the available niche space in response to predators, competitors, and prey due to their short life cycle (Dawe, Colbourne, and Drinkwater 2000). In the previous Assessment, short-finned squid were predicted to be positively impacted by climate change, although with high uncertainty (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Distribution and abundance of shortfin squid on the Northeast U.S. Continental Shelf is highly related to oceanographic conditions such as shelf-slope dynamics and positions of the shelf-slope front and Gulf Stream (Dawe et al. 2007). Dawe, Colbourne, and Drinkwater (2000) theorized based on catch data from Newfoundland that squid abundance may co-vary with that of its most important predators, competitors, and prey in an adaptive manner such that the population increases and expands when niche space becomes available. These shifts appear to result from high water temperatures, a negative North Atlantic Oscillation (NAO) index, and a southward shift in the Gulf Stream position. Greater catch per unit effort (CPUE) of shortfin squid has been associated with cooler bottom temperatures, timing and location of warm core rings in the winter and spring, and slope water/shelf slope front dynamics (Salois et al. 2023). These features may contribute to areas of high primary productivity, food sources, and transport mechanisms for larvae.

Life History Synopsis

The northern shortfin squid inhabits Northwest Atlantic shelf and slope waters from the Labrador Sea to the Florida Straits (Hendrickson and Holmes 2004). Spawning occurs year round, but peaks seasonally from October to June. Shortfin squid are semelparous; females release multiple egg capsules during a single spawning season, after which they die (Hendrickson and Holmes 2004). Laboratory studies indicate that hatching occurs in 16 days at 13°C, 12 days at 16°C, and 8 days at 21°C. In the laboratory, paralarvae enter a transitional stage at approximately 5.0 mm ML, followed by a juvenile stage at about 7.0 mm ML (O'Dor et al. 1982). The Gulf Stream and associated eddies are hypothesized to be the primary mechanisms for transport of larvae to northern latitudes and then inshore via the eddies (Trites 1983). Larvae have small

yolk-sacs, so they rely on rapid transportation to food-rich habitats (Hendrickson 2004). Larvae feed on a variety of prey items (fish, nematodes, euphausiids). Larvae and juveniles are abundant in the slope water of the Gulf Stream and in the adjacent upwelling regions at times of maximum production such as the spring plankton bloom (Hendrickson and Holmes 2004). As the initial biomass of the bloom is transferred up through various trophic levels, the squid grow at a pace which allows them to feed continuously at the level where biomass is peaking (O'Dor and Dawe 2013). Paralarvae, juveniles and adults of Illex exhibit diel vertical migrations to feed at night, and remain near the seabed during the day (Arkhipkin and Fedulov 1986). An ontogenetic shift in diet from a predominance of crustaceans to a predominance of fish and squid is evident as the squid mature (Dawe 1988). Larger, older Northern shortfin squid feed primarily on fish, but cannibalism of small individuals (most likely males) by larger females also occurs, particularly during autumn. Fish prey consists of the early life history stages of Atlantic cod, Arctic cod, redfish, sand lance, mackerel, Atlantic herring, haddock, and sculpin (Dawe 1988). Shortfin squid are important prey for seabirds and many fish species, including tunas, hakes, spiny dogfish, monkfish, and Atlantic cod (Hendrickson and Holmes 2004). Juveniles and adults make seasonal migrations off the continental shelf in the fall and back onto the shelf to spawning grounds in the Mid-Atlantic in the spring (Hendrickson and Holmes 2004). Shortfin squid are a short lived species, reaching maturity at 40-60 days and a maximum lifespan of less than one year (Hendrickson 2004). Adults are found at bottom temperatures between 3.5-15°C and surface temperatures greater than 20°C (Hendrickson and Holmes 2004). Shortfin squid in the Northwest Atlantic are managed by the Mid-Atlantic Fishery Management Council under the Mackerel, Squid, and Butterfish Fishery Management Plan. The most recent stock assessment occurred in 2025; management measures are in place but stock status is unknown as there are no reference points for this stock (Northeast Fisheries Science Center 2025).

Supplemental Scores

Exposure Timeseries

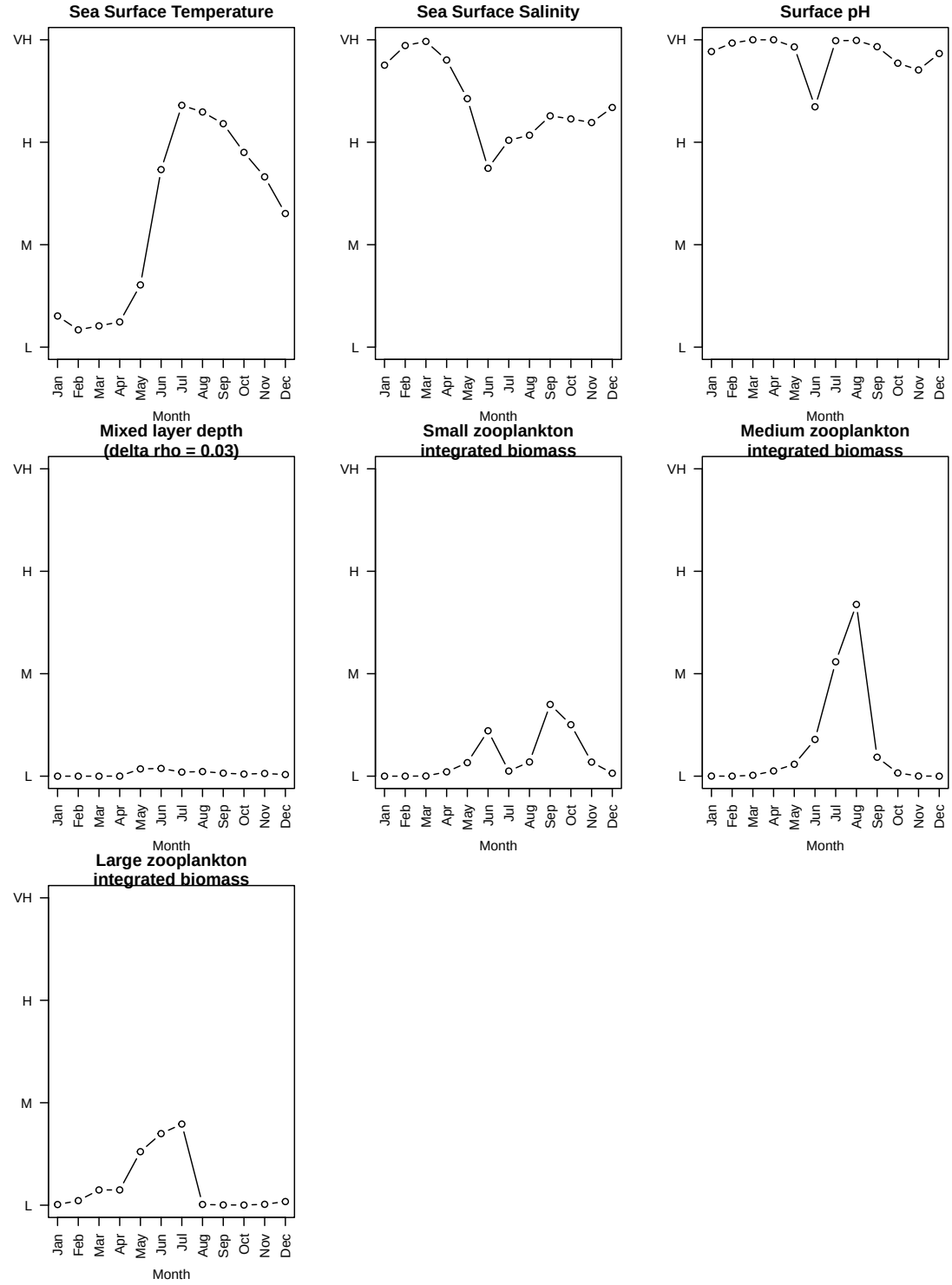


Figure 7: Monthly Timeseries of each Exposure Factor for Shortfin squid. Exposure was calculated between 2014-2023 and 2025-2035

There was little variation in exposure scores across months. Exposure scores were High for every month except June due to declines in sea surface salinity and pH exposure (Figure 8; Figure 7).

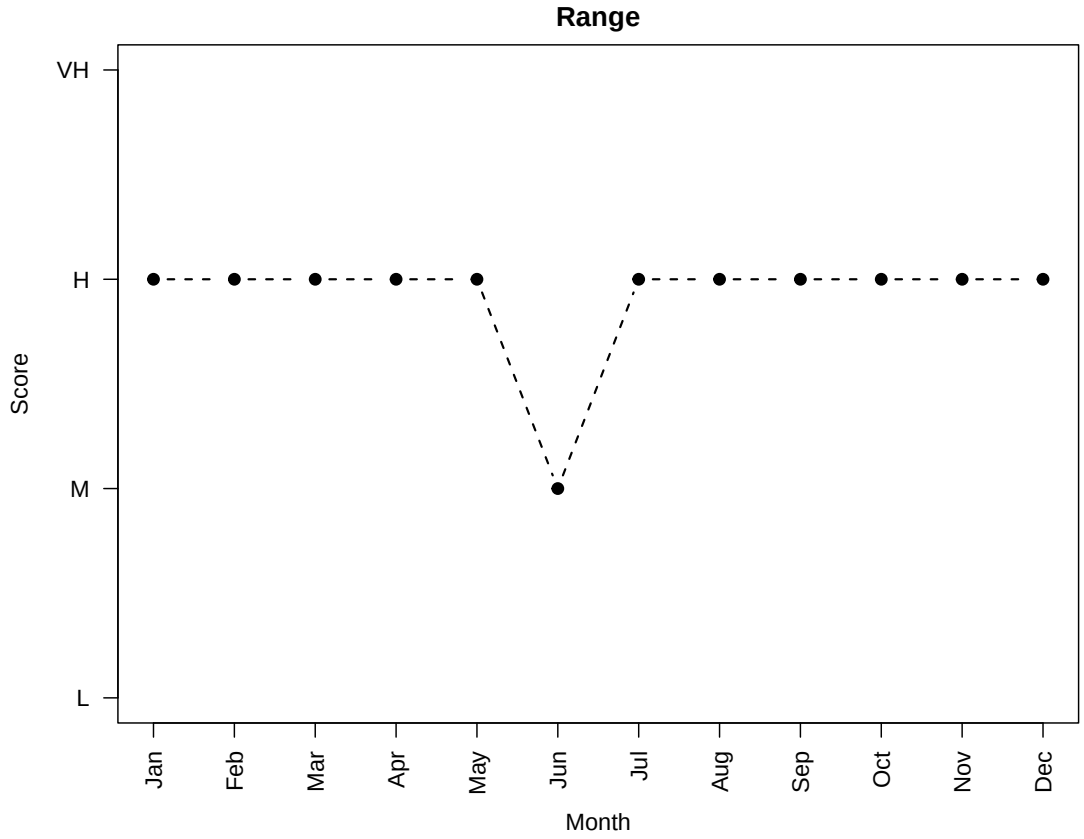


Figure 8: Timeseries of Total Exposure for Shortfin squid

Exposure due to Significant Factors

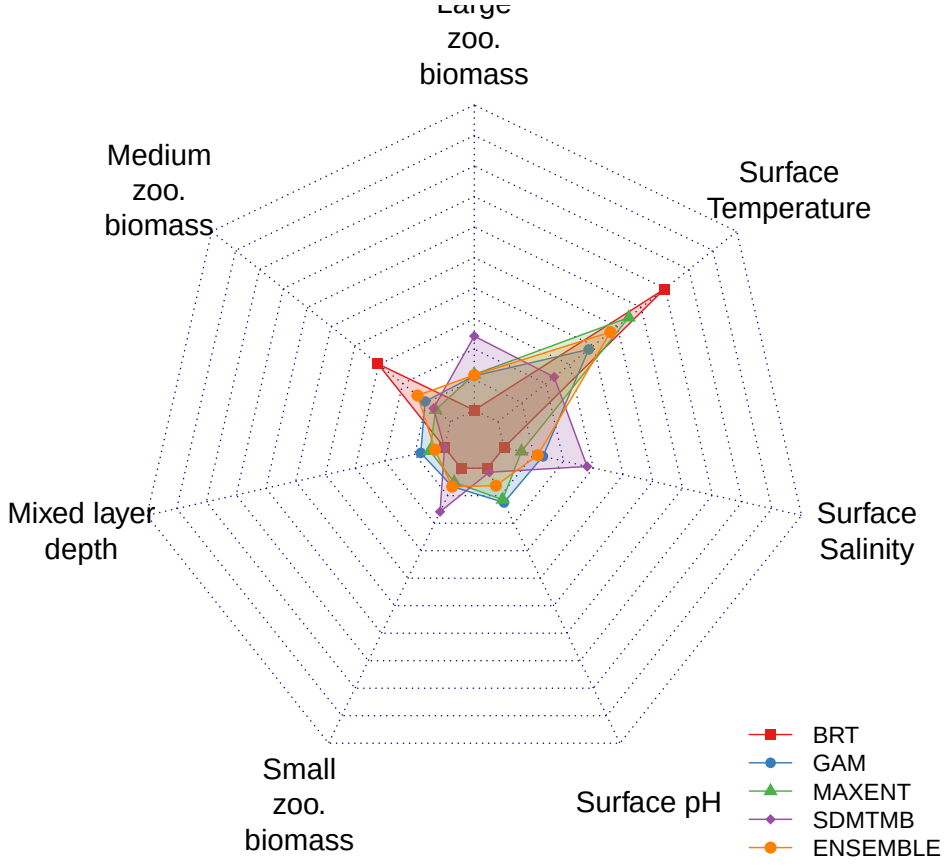


Figure 9: Dynamic Factors driving Shortfin squid distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Sea surface temperature, surface salinity, and medium and large zooplankton biomass were all important dynamic drivers of short fin squid distributions (Figure 9). Shifts in the Gulf Stream impact recruitment and shelf slope front dynamics (Dawe, Colbourne, and Drinkwater 2000) (O'Dor and Dawe 2013) (Hendrickson and Holmes 2004) (Salois et al. 2023). Short fin squid are also generalist predators feeding on crustaceans and copepods as juveniles and small fish as adults (Dawe 1988). Exposure was High throughout most of the shelf and Gulf of Maine, where areas of high surface temperature and salinity overlapped. Scores were Low in nearshore and shelf break areas (Figure 10). When only these factors were considered in the exposure score, mean exposure dropped to Moderate (1.93 +/- 0.91) (Figure 11).

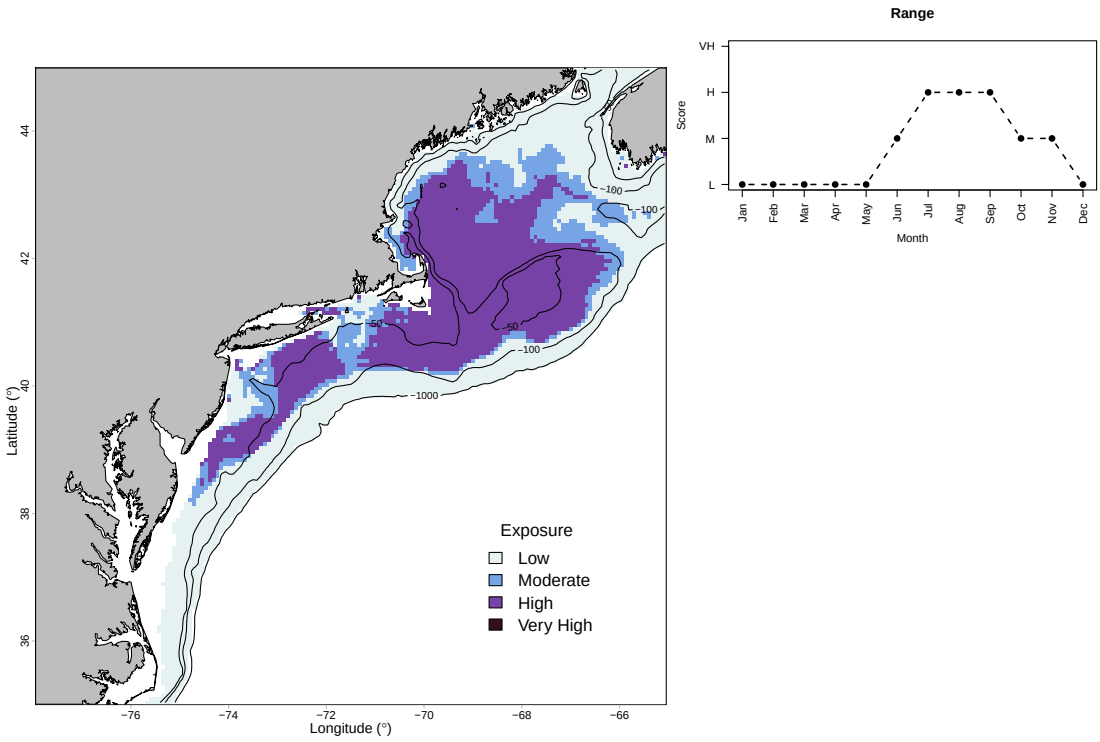


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Shortfin squid distributions

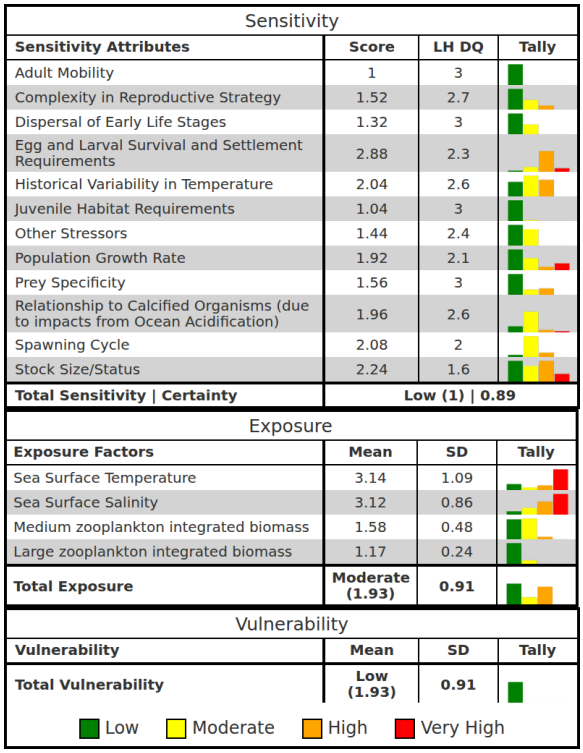


Figure 11: As in Figure 1 for Shortfin squid, but with only significant Exposure factors and the resulting Vulnerability Scores.

References

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